



Oral Surgery

Third stage

Lec. 17

Technique of forceps extraction

Dr. Mohammed Hassan
B.D.S., M.Sc., Ph.D. (Oral Medicine).

Zahraa Mohammed Alwan
MSc. periodontics

- Regardless of the tooth to be extracted in the oral cavity, **some common principles are applied to all dental extraction.**
- The **general steps** in forceps extraction procedure are:

1- Soft tissue retraction

- Before starting the application of the dental forceps, the gingival tissue surrounding the tooth should be reflected with blunt **probe or tweezers**, the neck of the tooth reflected with freed labially and lingually as far as the bony alveolar margin, for the following:

1. No laceration or tearing of the gingival tissues occur on extraction.
2. Reflection of the gingival tissue allows the surgeon to ensure that profound anesthesia is secured before starting extraction.
3. Also gingival retraction allow the beaks of forceps to be positioned more apically without interference or impingement of the gingival tissue

2- Handling of the forceps:

The forceps should be grasped by the palm of the right hand and the thumb finger is placed below the joint.

The **little** finger is placed **inside** the two handle of the forceps so that it can control the opening and closure of the handle and guide the forceps beaks on the root surface, and when the tooth is grasped the **little** finger is placed outside the handle.

Mandibular forceps



Placing the index finger inside the handles



Weak force and Low control



Greater force and Low control

Regarding the **upper molar** and **premolar** forceps, the curved side of the handle should rest on the palm of the hand



Position of the Operator

Operator should stand **in front** and slightly on the **right side** for **all teeth** extractions, except **right mandibular molar** teeth. He should stand behind at **11 o'clock** position for right mandibular molar extraction. This arrangement is for right handed operators, and the reverse for left handed operators.

Position of the operator for various extraction procedures.

<i>Extraction procedures</i>	<i>Position of the operator</i>
For all maxillary teeth, anterior mandibular teeth and teeth of the 3rd quadrant	The operator is to the front and right of the patient
For left-handed operator will stand on the left side	
For the teeth in the 4th quadrant	The operator stands behind and to the right of the patient
For left-handed operator will stand on the left side	

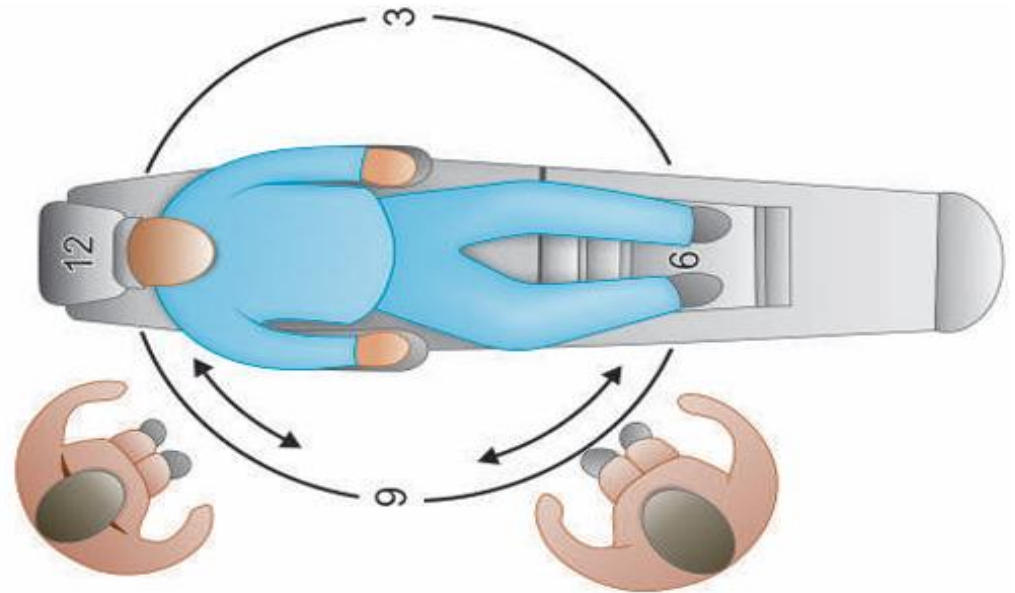


Fig. 18.14: Position of the operator for various extraction procedures.



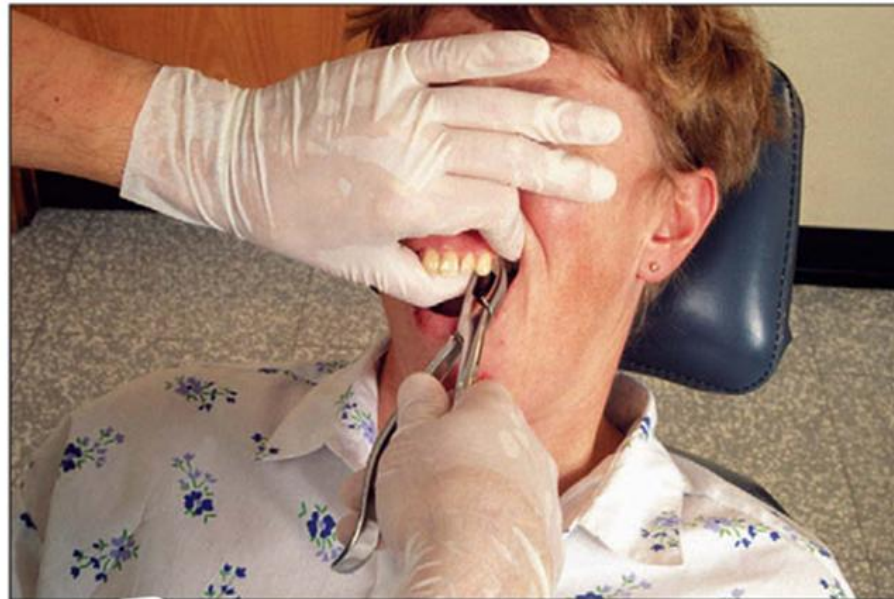
Position of operator for extraction of teeth from the left mandible.



Position of operator for extraction of teeth from the right mandible.

The supporting hand

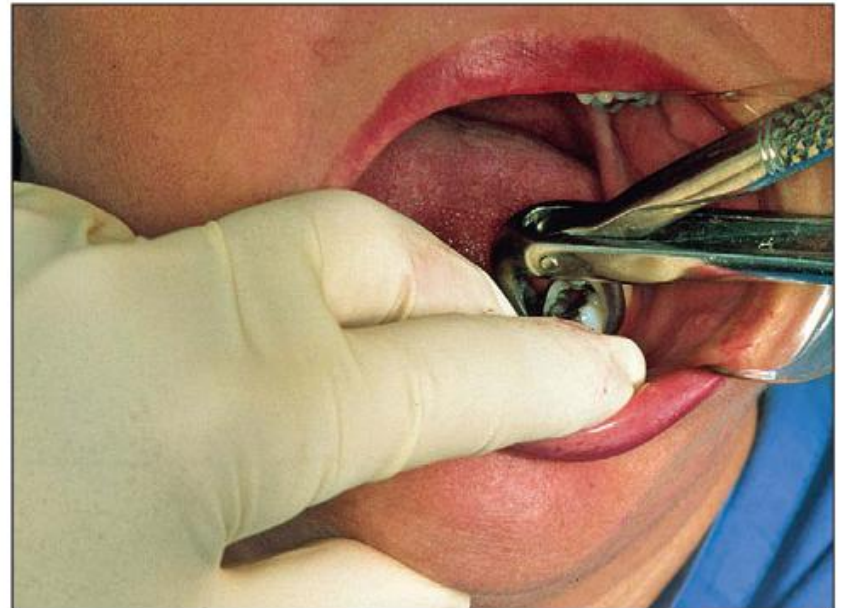
For the extraction of **left** maxillary teeth, the left **thumb** supports the palatal alveolar bone and **index finger** retracts the buccal tissues, while for **right** upper teeth, left hand **thumb retracts** the buccal tissues and **index** finger supports the palatal alveolar bone.



Supporting hand for maxillary extractions (thumb and index finger).

The supporting hand

For **lower left** teeth extractions, the **left thumb** supports the mandible and **index** finger retracts the buccal soft tissues, **middle** finger **controls** the tongue. For **lower right** teeth extractions, left hand **index finger** retracts the buccal tissues, **left thumb** controls the tongue and **other fingers** support the mandible.



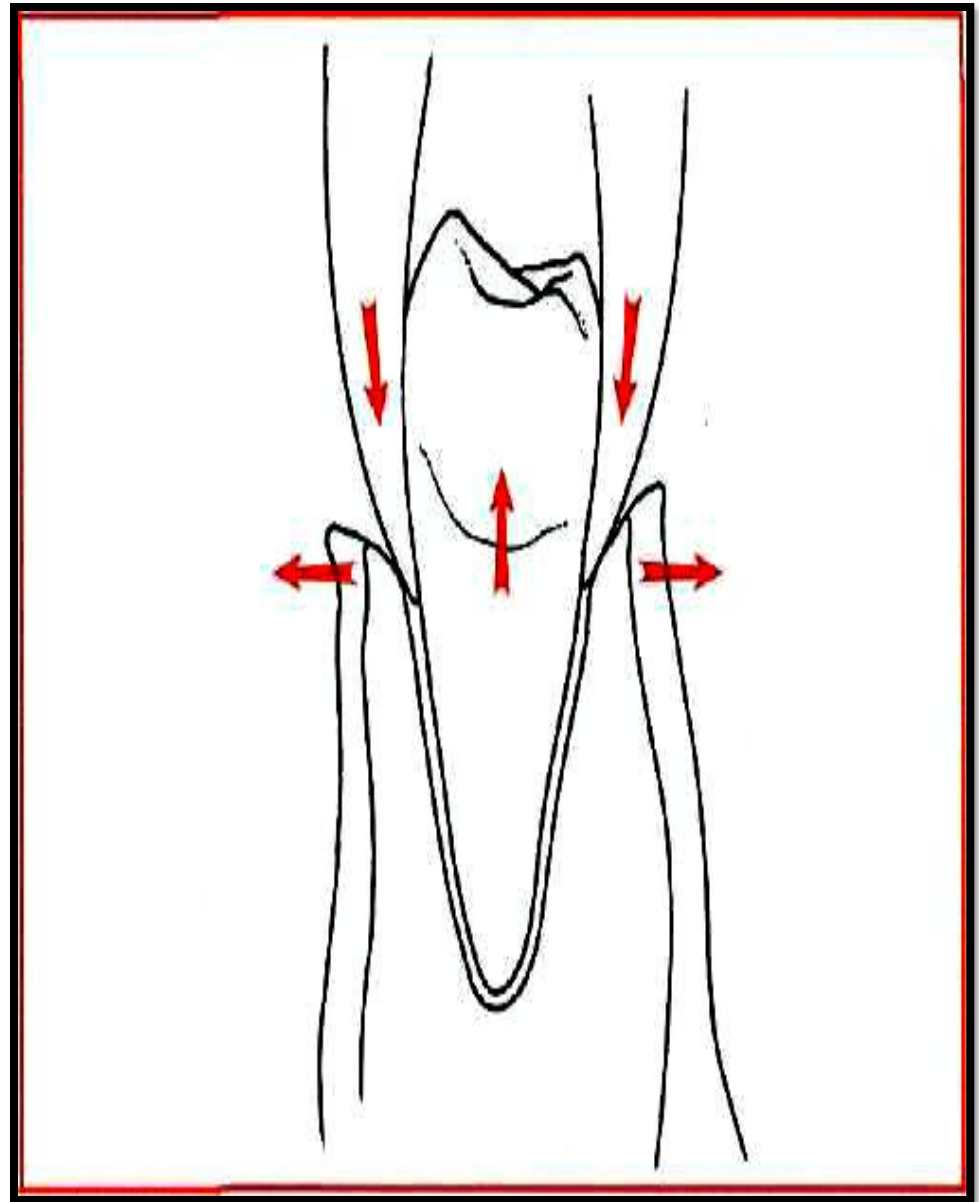
Supporting hand for mandibular extractions (thumb and two fingers).

3-The application of forceps blades to the (tooth grasp).

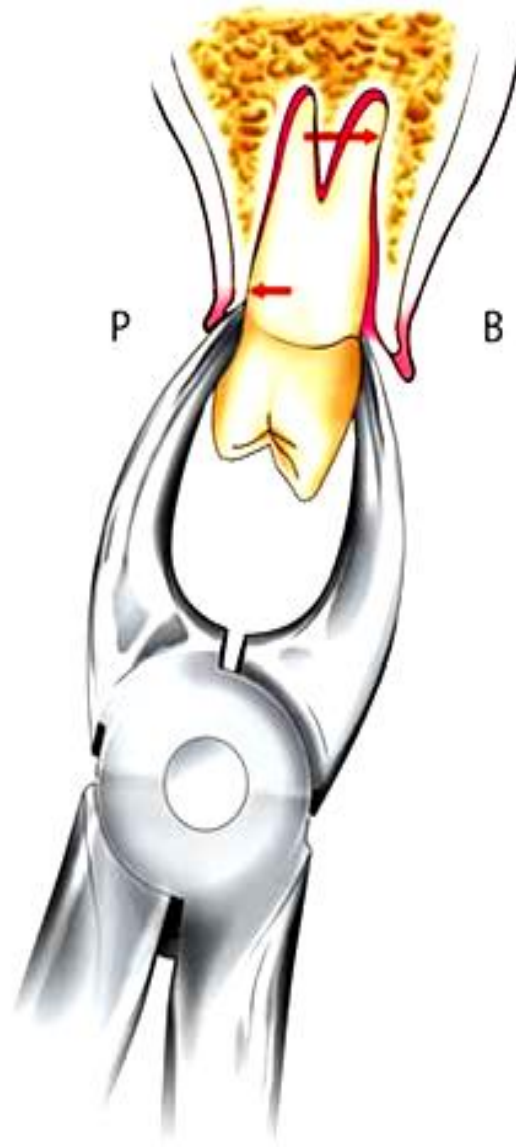
- After the selection of the proper forceps for the extraction of particular tooth. The forceps blades are applied on both labial(buccal) and lingual (palatal) surface the tooth, so that the blades are parallel to the long axis of the tooth to be extracted
- The blades of the forceps is moved on the tooth surface to grasp the tooth-root surface below the cemento-enamel junction.

- The tooth to be extracted is **grasped firmly**, the blades are not allowed to slide on the surface of the root during extraction movement.

The forceps
should be seated
with apical
pressure to expand
crestal bones and
to displace center
of rotation as
apically as
possible.



The extraction forceps blade should be applied to the carious side first, and the first movement made toward the caries.



4-displacement of the tooth from its socket:-

- This is performed by using the extraction movements.
- Important notes regarding the extraction movements:
 - ✓ The movement should be steady and with a reasonable force.
 - ✓ This pressure should be firm but not crushing, smooth and controlled not jerking.
 - ✓ The force should be held for several seconds to allow the bone time to expand.

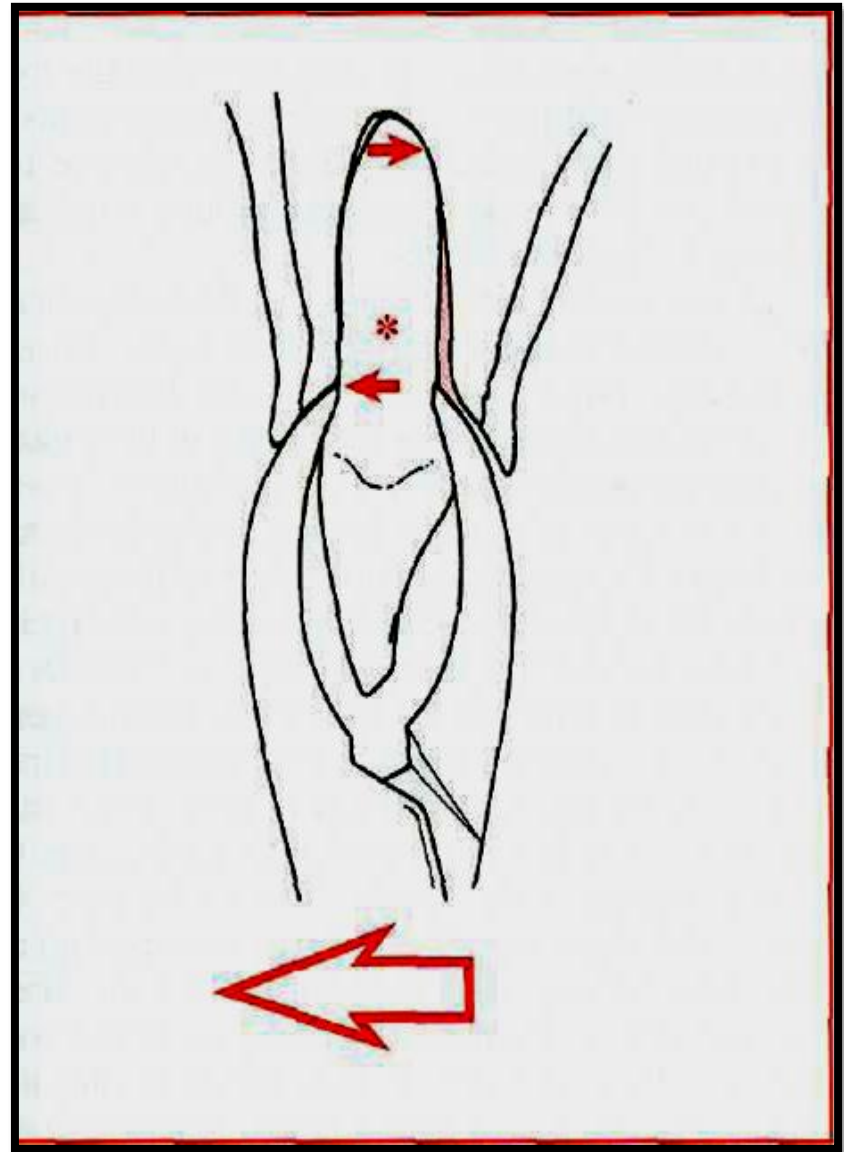
The extraction movements are:

1- Outward (buccal or labial) movement is the initial movement of all teeth except

❑ The lower second and third molar, where the buccal plate of bone reinforced by the external oblique ridge.

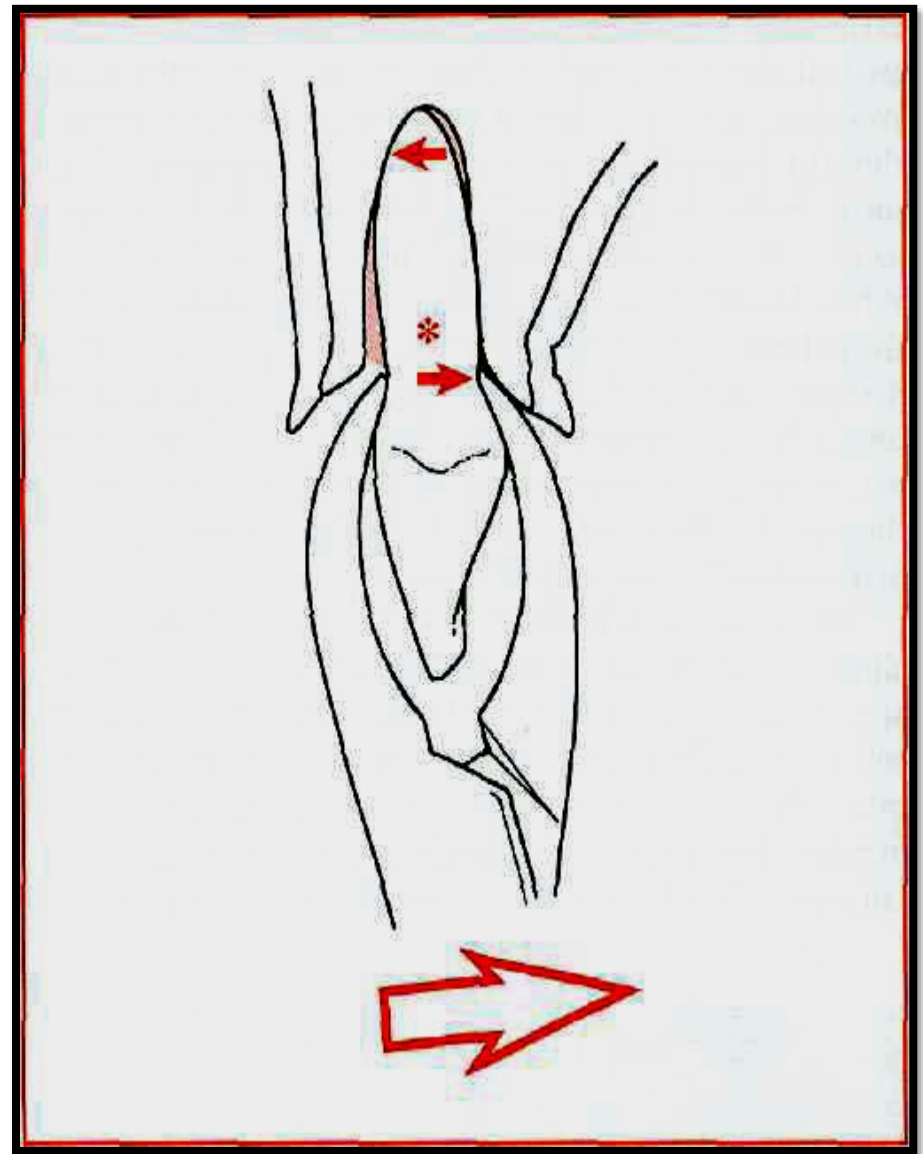
❑ upper central incisors and lower second premolars.

Buccal or labial pressure applied to tooth will **expand** the buccal cortical plate toward the crestal bone with some lingual expansion at apical end of the root.



2- Inward (lingual or palatal) movement is the initial movement during the extraction of the lower second and third molars.

Lingual or
palatal pressure
will expand
lingual cortical
plate at crestal
area and slightly
expand buccal
bone at apical
area.

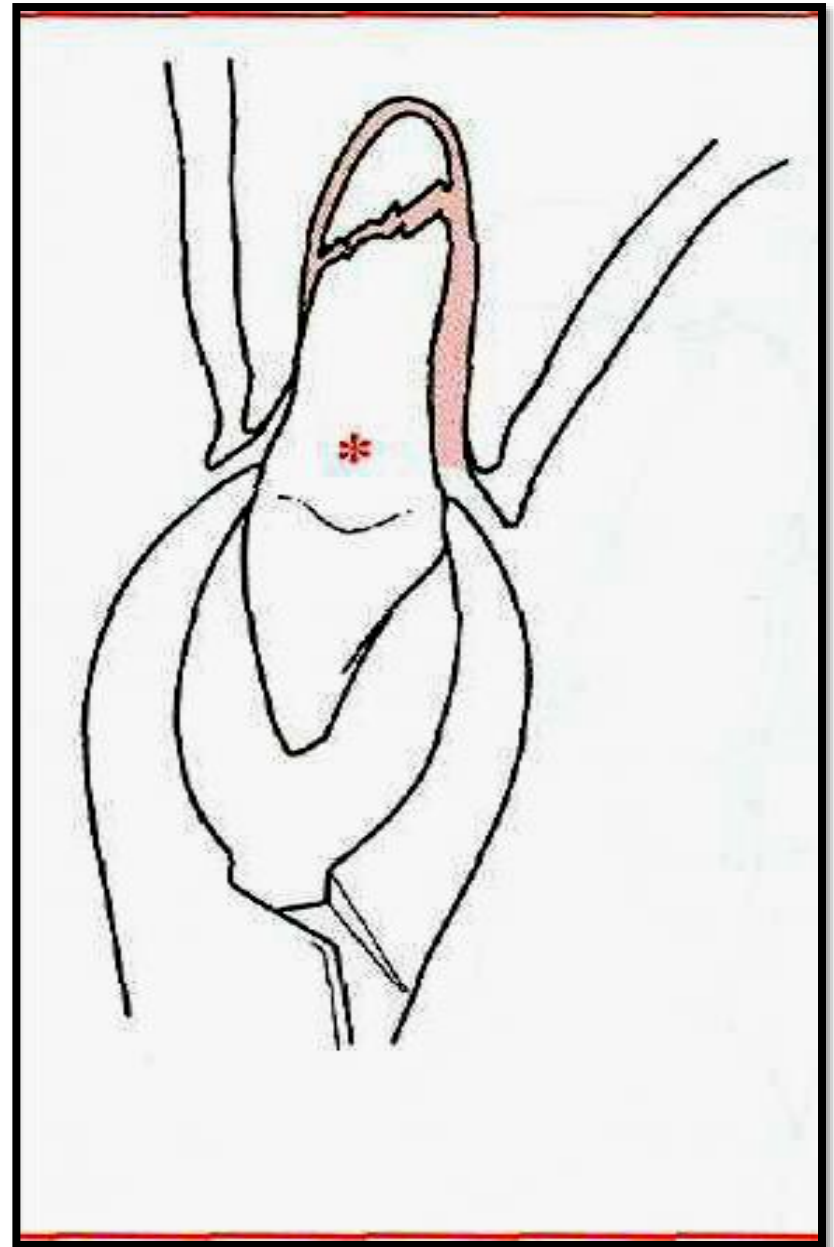


3- Rotatory movement: can be used as **primary movement** for extraction of teeth **with conical straight roots** such as the upper central incisors and lower second premolars.

If a **resistance is felt** in primary rotation, a **bucco-lingual movement** should be started.

If rotatory movement continued, **a spiral fracture** of the tooth root may occur.

If center of rotation is not far enough apically, it is too far occlusally, which results in excess movement of tooth apex, this excess motion of the root apex results in fracture of root apex.



Also rotatory movements are useful in completing the removal of teeth previously loosened by other means and this is called **secondary rotatory movement.**

Final movement

- Once the alveolar bone has expanded sufficiently and the tooth has **been luxated, a slight traction force, usually directed buccally**, can be used.
- Final movement is the movement by which the tooth is removed from its bony socket.
- It should be always **directed outward and occlusally** to avoid traumatizing the opposing tooth and preventing slipping of the tooth in the mouth which may be swallowed or aspirated by the patient.

Post-operative care of extraction wound:-

- 1- Examine the extracted tooth: **be sure that there is no fracture in the roots, keeping in mind the anatomical variations in number of roots** of each tooth, because some teeth has accessory roots
- 2- Always examine the **socket for any loose fragments of bone or roots** (pieces of enamel), foreign materials (e.g- calculus, amalgam particles, or pathology (e-g apical granuloma, polyp) all these should be removed by using curette or suction tip or tweezers.

- 3- Remove or smooth any sharp bone or projections of inter-radicular bone, then apply suture if necessary eg: there is laceration in the soft tissue.
- 4- The expanded bucco-lingual plate should be compressed or squeezed back to their original configuration or shape in order to reduce any distortion of the supporting tissue to re-establish the normal contour before extraction.

- 5- Make sure that the socket **is fill of blood** (to form blood clot).
- 6- Place a properly shaped and size gauze piece **over the socket**(and never put it into socket) in such a way that the patient close his mouth and it is not visible to gain initial control of hemorrhage.

Instructions to the patient

- 1- Keep biting over the gauze for at least 0.5- 1 hours and discharge after that but if hemostasis is not established, he can use another.
- 2- Do not spit during first few hours and use minimal talking and avoid violent exercise or activity to assist the formation of a firm clot in the socket, so the first few hours following extraction are recommended avoid the in the tongue tip inside the socket which lead to dislodgement of clot or disturb its formation.

- 3- use only fluid and soft diet for the few postoperative hours and not take solid or hard food for the rest of the day, and chew on a side opposite to the side of extraction

- 4- in case of continuous bleeding. Place a sterile gauze in place over the wound and keep it in place for an hour, if bleeding not stopped contact your dentist.
- 5- always prescribe the proper analgesic and antibiotic if necessary (e.g the presence of residual infection, diabetic patient, patient on immunosuppressive drugs,...etc